

Code Generation (ER → SQL)

Blaž Bone · Tilen Ožbot

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Supported ER Concepts — Entities & Relationships

Concept	T	G	Gen
Strong entity	✓	✓	✓
Weak entity	✓	✓	✓
Binary rel.	✓	✓	✓
Unary rel.	✓	✓	⊙
Ternary rel.	✓	✓	✓
N-ary (≥ 4)	✗	✗	⊙
Cardinality	✓	✓	⊙
Participation	✓	✓	✗
Roles	✓	✓	⊙

Supported ER Concepts — Attributes & EER

Concept	T	G	Gen
Entity attr.	✓	✓	✓
Rel. attr.	✓	◐	✓
Keys	✓	✓	✓
Datatypes	✓	✓	✓
Composite	✗	✗	✗
Multivalued	✓	✓	✓
Derived	✓	✓	✓
Generalization	✓	✓	✓

T = textual · G = graphical · Gen = generator · ✓ supported · ◐ partial · ✗ unsupported

Assumption 1 — Explicit Relational Artifacts

- The generator uses deterministic ER → relational mapping rules
- Strong entities become tables
- Relationships become bridge tables
- Weak entities copy the owner key and add their `partial_key`
- Multivalued attributes become side tables
- Inheritance becomes a child table with a primary-key foreign key to the parent

Why:

- SQL output stays easy to inspect
- Golden fixtures stay readable
- Generated schemas can be validated with real database engines

Assumption 2 — Constraints Where SQL Can Express Them

- Primary keys are generated from `key` attributes
- Weak entity identity is generated from owner key + `partial_key`
- Relationship participants become foreign keys
- Binary max-one cardinalities become `UNIQUE` constraints
- Weak ownership and composition use `ON DELETE CASCADE`
- Optional attributes become nullable columns

Boundary:

- Mandatory participation is not enforced in generated SQL
- We do not generate triggers or assertions
- Some ER semantics remain modeling-level information

Assumption 3 — Dialects Normalize Types, Not Semantics

Example input:

```
score: DOUBLE  
photo: BLOB  
bio: CLOB  
birthday: DATETIME
```

ER datatype	PostgreSQL	MySQL
DOUBLE	DOUBLE PRECISION	DOUBLE
BLOB	BYTEA	BLOB
CLOB	TEXT	TEXT
DATETIME	TIMESTAMP	TIMESTAMP
BOOLEAN	BOOLEAN	BOOLEAN